Smaller explicit Diophantine forms of ZFC consistency

Mixed radices, cofinal bases, and norm cancellations

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Abstract

We compare the seven current ProveIt constructions, numbered 24–30, and combine features that their separate presentations leave uncombined. The main new variable reduction gives one explicitly specified polynomial in sixteen unrestricted integers. Its degree is $813056 \cdot 5^{32} + 10024$, and its delivered arithmetic circuit uses 1865 binary operations. A seventeen-integer alternative has degree $406528 \cdot 5^{32} + 27512$ and 1874 operations. A cofinal family of square radices removes one positivity witness; a smaller signed separating weight also lowers the degree. For positive unknowns we obtain a twelve-variable equation of degree 1065, and separately reduce the 1081-degree equation from 111 to 108 operations. Three coordinate shifts give a 31-variable quadratic system using 62 operations, or a quartic equation using 127. All families can use a new, fully materialized four-coefficient tuple with $B = 2^{1064921}$, a substantial reduction from $2^{1953125}$. We give ordinary mathematical proofs, literal arithmetic circuits, coefficient generators, and independent finite audits. The equivalences are formalizable in ZFC; the new reductions are not claimed to be new end-to-end Lean or Rocq certificates, globally minimal presentations, or proofs of ZFC consistency.

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1 Outcome, comparison, and interpretation

Write $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{N}_{>0} = \{1, 2, \ldots\}$. For every construction delivered here, with its explicitly stated domain D, the intended theorem is

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in D : P(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{1}$$

For a system, $P = 0$ in this display means that all listed equations hold. The consistency predicate means that no finite derivation of contradiction exists in the fixed effective presentation of ZFC used by the K215 endpoint. The proof establishes the equivalent positive statement: a solution exists if and only if a finite ZFC refutation exists.

There is no single ordering that makes an equation with fewer variables, another with smaller degree, and a third with fewer arithmetic operations all "simplest." We therefore retain the useful new points on these tradeoffs and identify their exact gains. An operation is one binary $+$, $-$, or multiplication, including multiplication by a fixed numeral. Reuse and equality comparisons are free. Construction of fixed integers is accounted for separately from evaluation of a polynomial at an input. The degree counts quantified variables only: coefficient ports have degree zero. These conventions agree with the frozen repository catalogue.

Criterion	Frozen predecessor	This report
Unrestricted integer count	17 variables, 20607 operations	16 variables, 1865 operations
17-integer alternative	Degree $424296 \cdot 5^{33} + 53824$	Degree $406528 \cdot 5^{32} + 27512$
Single-equation degree in 12 positive variables	Degree 1069, 123 operations	Degree 1065, 118 operations
Single-equation cost in 12 positive variables	Degree 1081, 111 operations	Degree 1081, 108 operations
Quadratic system in 31 positive variables	22 equations, 65 operations	22 equations, 62 operations
31-positive-variable quartic	130 operations	127 operations
Smallest materialized base	$B = 2^{1953125}$	$B = 2^{1064921}$
Four coefficient payloads	977220 raw bytes	533099 raw bytes

The sixteen-integer polynomial has degree $813056 \cdot 5^{32} + 10024$, which is also lower than the predecessor's seventeen-integer degree. It is still enormous. For a readable low-degree presentation, the 22-equation quadratic system is the more useful choice. For a compact direct positive-domain circuit, the 108-operation equation is preferable. For the fewest positive variables, the existing Jones nine-variable route remains relevant: this report does not improve the count nine. The established 67-operation single equation and 59-operation higher-degree systems are also not beaten in their own operation-count categories.

The comparisons concern explicit constructions in the inspected repository. They are not assertions of universal optimality. In particular, Sun's September 2026 preprint states an effective ten-integer representation theorem for every recursively enumerable set [8]. That theorem gives a smaller general existence bound than sixteen. Its different source coding has not been instantiated into a fixed K215 polynomial in this delivery; calling the sixteen-integer circuit the globally smallest known integer equation would therefore be unjustified.

1.1 Sources and the frozen snapshot

The source repository is https://github.com/VladimirReshetnikov/ProveIt. All comparisons and source copies in this archive use commit

`8c6eec9d19ef80538c8e61ff6e0180040583446a`

in the directory

`SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/con-zfc-diophantine-forms/`

The main catalogue at that commit already incorporates its audit of the fourth round of submissions. We compare those audited files, rather than treating a historical summary or a package's claim about its predecessors as current evidence.

The useful contributions are complementary. Package 24 provides a tight separating weight and a padding argument for general odd radix dimensions. Package 25 provides mixed-radix local decoding, compact clause bands, and population adjustment in fields that are identically zero. Package 30 provides a trace/product norm evaluator and the 31-variable quartic. Package 29 provides a signed Jones finalizer with a stack of positivity certificates. Combining the last two is especially effective: the signed construction had not yet used the smaller quartic or the compact evaluator.

2 The proof interfaces and what is verified

The inherited proofs used here have four precise roles.

1. The literal 215-state machine on the blank tape halts if and only if ZFC has a finite refutation. The pinned Lean endpoint states this as `KEndpoint215.blank_halts_iff_zfc_refutation`; its statement is included in the archive. We use the corresponding ordinary mathematical calibration, rather than an assumption that ZFC proves the soundness of Lean as a whole.
2. The fixed 943-tile language represents finite blank computations. A periodic tiling with a Start marker and no Forbidden tile yields a halting computation. Conversely a finite halt can be padded to arbitrarily large independent widths and heights. These are the exact semantic properties needed when we change the numerical layout.
3. The native-window and Pell compiler translates that language into the direct polynomial interface. Its soundness is for arbitrary solutions, and its completeness constructs positive witnesses. The canonical witnesses additionally have square scale w, odd scale s, and an odd strong-Pell multiplier where the signed reductions use them. The materialized tuple must satisfy this whole interface; modular identities of four unrelated integers would not suffice.
4. The Jones reduction translates a normalized quartic into its eight-coordinate ratio predicate, or into a nine-variable equation over $\mathbb{N}$. The signed finalizer uses that particular predicate and its cofinal-radix completeness proof. It is not a generic conversion of an arbitrary nine-variable polynomial into sixteen integer variables.

The new coefficient adapter is proved in Section 3. The new arithmetic reductions and the signed construction are proved in the later sections. These proofs, together with the cited inherited lemmas, give (1) by ordinary mathematics formalizable in ZFC. The archive includes the relevant inherited proof notes, literal source data, and source locations so that this dependency is reviewable. The native compiler and new finalizers remain outside the existing end-to-end proof-assistant certificates.

Finite checking has a distinct, narrower role. It verifies the literal tile list, numerical coefficients, convolution identities, size bounds, operation counts, coordinate identities, norm identities, and degree certificates. It does not search for a counterexample to consistency, simulate an unbounded machine run, or establish the universal compiler theorem by testing finitely many cases.

3 A smaller complete coefficient tuple

The new backend combines the mixed-radix local decoder and compact bands of package 25 with the general odd-factor Boolean padding argument of package 24. Neither original numerical tuple is reused piecemeal. All four ports are constructed together and then audited together.

3.1 The finite language and its literal encoding

The file `coefficients/k215.tm` contains the 430 literal actions of the 215-state machine. Its SHA-256 is

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090.`

The generator never runs this machine. It reads the finite transition table, constructs the same 943 ordered tiles as package 25, and checks the canonical tile-list hash. A tile has west, north, east, and south edge labels. The language contains 430 transition tiles, two halted-head loops, two unchanged-bit tiles, 398 right-arrival tiles, 90 left-arrival tiles, four finish tiles, four initialization tiles, three reset tiles, nine seam tiles, and one Forbidden tile. Start is placed first, Forbidden second, and the remaining tiles are lexicographically ordered.

For clarity, the finite semantic contract retained from the source is as follows. A marked periodic plane forces initialization between reset boundaries. The directed signal colors update exactly one head at each time step and cannot cross a boundary. A new reset row can occur only after a halted head has appeared. Thus a component containing Start contains a finite blank halting computation. Conversely, a finite halt admits arbitrarily wide blank margins and arbitrarily many halted padding rows; the seam tiles close these to a helical periodic tableau. Its width and height can be chosen independently among all sufficiently large integers, in particular among powers of the prime used below.

Let

$$b_c = 121, \qquad L_c = 8801, \qquad d_c = b_c L_c = 1064921, \qquad \rho = 2^{121}, \qquad B = \rho^{8801}.$$

The native support is

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1032\} \\ & \cup \{1063, 1064, 1095, 1096, 2195, 3296\}.\end{aligned}$$

It has 989 elements. The two dummy positions are 1063 and 1064; the first permits the harmless binary variation used for index control. For the four edge slots use copy offsets $o_0 = 1023, o_1 = 1013, o_2 = 1003, o_3 = 983$. Put $w_j = 2^{10+2j}$, for $0 \le j < 44$. If tile i has labels $x_{i,t}$, its clause coefficient is

$$c_i = 1 + \sum_{t=0}^{3} \sum_{j=0}^{9} \mathrm{bit}_j(x_{i,t} + 512) w_{10t+j} + \sum_{j=40}^{43} w_j.$$

At the copy position $o_t + j$, set $c_{o_t+j} = w_{10t+j}$. At the anchors $1095, 1096, 2195, 3296$, set the coefficients to $w_{40}, w_{41}, w_{42}, w_{43}$, respectively. Both dummy coefficients are zero. The clause mask is

$$\mu = 1022 + \sum_{j=0}^{43} w_j.$$

Its bit length is 97 and its population is 53.

The two clause targets are $T_1 = 4393, T_2 = 5495$ and the spatial displacement is 42. Define the fixed integers

$$D_C = \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2200} + \sum_{e\in\mathcal{E}} c_e(\rho^{4393-e} + \rho^{5495-e}),$$
$$D_R = \rho^{42},$$
$$M_C = B - 1 - \sum_{\substack{e\in\mathcal{E}\\ e\neq 1}} \rho^e - 2\rho^{1063},$$
$$M_{F,\text{native}} = 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9} \rho^{1013+j} + \sum_{j=0}^{9} \rho^{1065+j} + \rho^{2195} + \rho^{3296} + (2^{119} - 1) \sum_{e=8792}^{8798} \rho^e + (2^{27} - 1)\rho^{8799}.$$

Every exponent is nonnegative. The four actual ports are

$$\boxed{\beta = B - 1, \quad K_0 = D_C + BD_R, \quad M_C, \quad M_F = M_{F,\text{native}} + B - 1.} \tag{2}$$

In particular M_F is already shifted. The unshifted mask is not the fourth arithmetic input.

3.2 Decoder, separation, and carries

The low ten bits of a clause sum count the selected tiles. Before any decoding there are at most 943 selectors, and all higher coefficients are multiples of 1024. The mask 1022 therefore forces the selector count to be either zero or one. Conditioned on this conclusion, each of the 44 remaining radix-four digits is a sum of two Boolean values. It has no carry and its masked parity forces the two values to be equal. This gives both the edge-copy constraints and the anchor constraints. The order of these arguments matters: carry-freeness of the radix-four digits is used only after the selector count has been recovered.

The overlap tests are unchanged. At position $1013 + j$, the bare shift 30 pairs the south copy $983 + j$ with the temporal north copy $1013 + j$. At position $1065 + j$, the bare shift 62 pairs the east copy $1003 + j$ with the spatial west copy $1023 + j$ shifted by 42. The two anchor tests at 2195 and 3296 pair copies at 1095 and 1096, respectively, with their temporal copies. At a clause target, the reflected coefficient from each support position is exactly c_e, and all other possible contributions vanish.

These are identities on every native basis vector, not observations about selected example tableaux. There are 69230 such finite identities in the delivered audit. The two further support identities are

$$\{(4393 - e) \bmod 8801 : e \in \mathcal{E}\} \cap \{(5495 - e) \bmod 8801 : e \in \mathcal{E}\} = \varnothing,$$
$$\{(2195 - e) \bmod 8801 : e \in \mathcal{E}\} \cap \{(3296 - e) \bmod 8801 : e \in \mathcal{E}\} = \{0\}.$$

The first preserves a clean clause band under arbitrary rotations; the second forces whole-cell synchronization. The source's two-anchor parity argument first excludes subdigit offsets, including the one possible upper-dummy spill. The displayed intersection then excludes all remaining nonzero offsets. Thus this part of soundness uses no power-of-five assumption on the new dimensions.

Put

$$\mathcal{M} = (989 + 2)\left(2\sum_{e\in\mathcal{E}} c_e + 6\right).$$

Its bit length is 118, while that of $4\mathcal{M}+8$ is 120. The complete conservative threshold is

$$\max(4\mathcal{M}+8, 2\mu+4, 2(2^{119}-1)+4, 16) = 2^{120}+2 < \rho.$$

This bound applies before selector extraction, including all allowed dummy values. It preserves the no-carry and positive raw-slack hypotheses of the source compiler.

The largest unshifted native product position is $5495+3296=8791$. The padding masks occupy only positions 8792 through 8799, and $8799 < 8801$. They therefore test fields that are structurally zero in every canonical tableau, even after dummy-bit changes. Their population is $7 \cdot 119 + 27 = 860$. Consequently

$$\mathrm{pc}(M_C) = d_c - 989, \qquad \mathrm{pc}(M_{F,\mathrm{native}}) = 2 \cdot 53 + 22 + 1 + 860 = 989.$$

Both masks are strictly between zero and $B-1$, and

$$M_C \equiv 2 \pmod 4, \qquad M_{F,\mathrm{native}} \equiv 4 \pmod 8.$$

The population identity and residues are verified on the materialized integers as well as on their finite recipes.

3.3 Completeness for the new odd dimensions

It would be incorrect merely to use the new dimensions in the old power-of-five completeness argument. Instead factor

$$d_c = 11^2 \cdot 13 \cdot 677$$

and choose the independently verified prime $\lambda = 2341$ and the step $k_c = 2340 = \lambda - 1$. The exact order data are

Modulus	121	13	677	2341
Order of B	10	12	52	60

Thus $\mathrm{ord}_{d_c}(B) = 780$ and k_c is a common multiple of all orders. It is deliberately not claimed to be their least common multiple. Also

$$B^{2340} \bmod 2341^2 = 2556373 = 1 + 1092 \cdot 2341.$$

The valuation $v_{2341}(B^{2340}-1)$ is therefore exactly one. Writing $H_0 = 1 + BD_C + B^2(D_R+1)$, the unit check is

p	$B \bmod p$	$D_C \bmod p$	$(B^2-1) \bmod p$	$H_0 \bmod p$
11	2	5	3	9
13	6	1	9	7
677	611	38	293	654
2341	2100	2051	1896	1225

Choose sufficiently large independent tableau width w_c and height h_c that are powers of λ, and put $N_c = w_c h_c = \lambda^n$. Require $N_c > d_c\lambda^2$ and $h_c > \lambda$; padding a fixed finite halt permits these requirements. Put $q = B^{N_c}$. Since $\lambda \equiv 1$ modulo the relevant orders,

$$q \equiv B^{w_c} \equiv B \pmod{d_c\lambda}.$$

Changing a free low dummy in cell j changes the actual packed Pell index by $-\Gamma B^j$, where

$$\Gamma = \rho^{1063}(q^2-1)[1+q(D_C+BD_R+B^{w_c})].$$

Every prime factor of d_cN_c is in the unit table. Thus $\gcd(\Gamma, d_cN_c)=1$.

Lemma 3.1 (Boolean coverage of the required modulus)**.** *Put $T = N_c/\lambda$. Every residue modulo d_cN_c is a sum of a Boolean subset of*

$$\{B^{2+k_cj} : 0 \le j < T\} \cup \{B^3\}.$$

The chosen cells and their required spatial and temporal translates can all be kept nonwrapping.

Proof. Let $C_0 = d_c\lambda$. Then $d_cN_c = C_0T$ and $T > C_0$. The valuation calculation and the binomial lifting identity show that B^{k_c} has order T modulo λ^n and order one modulo d_c. Its first T powers are exactly the T residues $1 + C_0z$ modulo C_0T. This follows either by the Chinese remainder theorem and cardinality or directly from the order calculation.

Multiply the target by B^{-2}. Choose $\varepsilon \in \{0,1\}$ so that its difference from εB is nonzero modulo λ. Its least residue a modulo C_0 then satisfies $1 \le a < C_0 < T$ and $\gcd(a,T)=1$. Write the remaining target as $a + C_0u$. Solve

$$az_0 + \frac{a(a-1)}{2} \equiv u \pmod{T}.$$

The a distinct orbit entries with additive coordinates $z_0, z_0+1, \dots, z_0+a-1$ have the desired sum. Include B^3 exactly when $\varepsilon = 1$. All selections are Boolean.

The largest available cell position is $2+(\lambda-1)(N_c/\lambda-1) = N_c - N_c/\lambda - \lambda + 3$. As $h_c > \lambda$, the width w_c is less than N_c/λ. Translations by one and by w_c therefore remain below N_c. The additional cell 3 is distinct and also safe. □

Start with a canonical tableau whose dummy bits are zero, and let its packed index be R_0. Apply the lemma to $(R_0 - d_cw_c)\Gamma^{-1}$ modulo d_cN_c. The modified tableau has $R \equiv d_cw_c \pmod{d_cN_c}$. This is exactly the temporal rotation demanded by $X = 2^R$. The changes affect no selected tile, matched edge, Start marker, zero padding field, or carry estimate. The inherited Pell completeness argument now supplies all positive witnesses.

3.4 Canonical parity and the signed interfaces

Let $u_c = d_cN_c$, so $q = 2^{u_c}$. All dimensions are odd. The canonical Start digit and forbidden selector imply that the native code C is 1 modulo 4. With

$$S = (C-1) + qF, \qquad T_m = (M_CJ+1) + q(M_{F,\text{native}}J - 1), \qquad M = q^2,$$

the source packing identity is

$$R = (M-S)(M-1) + T_m = (M-1-S)M + (S+T_m).$$

The native masks give disjoint binary support of S, T_m; the exact population balance gives $\mathrm{pc}(T_m) = u_c + 2$. The upper complement then proves $\mathrm{pc}(R) = 3u_c + 2$. The low residues also give $R \equiv 3 \pmod 4$.

The canonical scale $w = 2^{R-u_c}$ is consequently an integer square. For the even main Pell parameter, its coordinate c at the odd rank R is odd; choosing the strong rank cR makes the normalized multiplier

odd, and its ordinary discriminant-scaled version is odd as well. Finally, the inherited half-binomial valuation identity $v_2(Y) = \mathrm{pc}(R) - 2 = 3u_c$ shows that $s = Y/q^3$ is odd. Thus the new tuple supplies the full canonical odd and square properties, in addition to the unsigned compiler equivalence.

Theorem 3.2 (Complete replacement of the four ports)**.** *The tuple* (2) *satisfies the same blank-K215 native-window/Pell interface as the Wang943 source, including its canonical odd and square witnesses. It can therefore be used as a whole tuple in the compatible direct forms, the ordinary quartic, and the Jones and signed constructions based on that quartic.*

3.5 Materialized size and reproducibility

The archive contains the actual four integers, both as raw minimal unsigned little-endian byte strings and as gzip files. Their sizes are

Port	Bit length	Raw bytes	Gzip bytes
β	1064921	133116	164
K_0	1070004	133751	9794
M_C	1064921	133116	264
M_F	1064922	133116	263
Total		533099	10485

The exponent of B is about 45.48% smaller than the preceding minimum; the raw payload is about 45.45% smaller. Compression is reported only as a transport size, not as coefficient height. These are four actual integers, fixed by a terminating finite recipe independently of whether the machine halts. The much larger Jones index coefficients are separately specified by exact finite recipes in the signed construction.

The builder and the independent audit use only Python's standard library. The latter reconstructs the tuple without importing the builder and checks every port against the delivered bytes. Both the generator and audit can be rerun without network access.

4 A 62-operation quadratic system and its quartic

For a presentation with small degree, the most readable new result is a system of 22 quadratic equations in 31 strictly positive integers. Three coordinated changes of coordinates reduce package 30's system from 65 operations to 62. Its sum of squares is a single quartic using 127 operations, compared with 130 before the changes.

The 31 variables are

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, T, c, a, X, D, C_2, S, F_2, V, g, I, B_a, \Delta.$$

All are strictly positive. Use the four fixed coefficients (2) and the affine abbreviations

$$q = \beta J + 1, \qquad k = \eta + \zeta, \qquad H = 4a + 3, \qquad C = q - F - \alpha.$$

The equations are

$$S = q^2, \qquad N = qS, \qquad X = wq, \tag{3}$$
$$a = E + sN, \qquad E = X(a - E), \qquad c = k(a - E) + \eta, \tag{4}$$
$$T = E(c - \eta), \qquad \tau^2 = T(T + k) + 1, \tag{5}$$
$$D = X + ac + \gamma H, \qquad \Delta = a^2 + H, \tag{6}$$
$$C_2 = c^2, \qquad D^2 = \Delta C_2 + 1, \tag{7}$$
$$k = jc - V + 1 + hE. \tag{8}$$

The two packing and transport equations are

$$jc - V = (q - F)(N - q) - C(S - 1) + (M_C + qM_F)J, \tag{9}$$
$$(K_0 + w)C = F + z(q - 1). \tag{10}$$

The remaining seven equations are

$$F_2 = f^2, \qquad V = of - c, \qquad g = y^2 - V^2, \tag{11}$$
$$I = iC_2, \qquad B_a = I^2, \qquad B_a = \Delta(F_2 - 1), \tag{12}$$
$$y^2 - B_a g = 1. \tag{13}$$

Since every abbreviation is affine and every displayed product has at most two affine factors, every residual has total degree at most two. Writing the 22 residuals as r_j, the single polynomial

$$Q_{31} = \sum_{j=1}^{22} r_j^2 \tag{14}$$

has degree exactly four: the coefficient of τ^4 is one. Over the stated positive domain, $Q_{31} = 0$ is equivalent to all 22 equations.

4.1 The three shifts and the positive zero sets

The predecessor coordinates being changed are denoted by $L_{\text{old}}, P_{\text{old}}, z_{\text{old}}$. The exact substitutions are

$$L_{\text{old}} = S - 1, \qquad P_{\text{old}} = T + k, \qquad z_{\text{old}} = z + 1.$$

All other variables are untouched.

The old equations for L_{old} and N become $S = q^2$ and $N = qS$. In the packing equation L_{old} becomes $S - 1$. This removes two gates and introduces one, for a net saving of one. The old first-root definition was $P_{\text{old}} = E(c - \eta) + k$; after the shift it is $T = E(c - \eta)$. The first norm becomes $\tau^2 = T(T + k) + 1$. This saves a second gate. Finally the transport identity

$$(K_0 + w)C + (q - F) - (z + 1)(q - 1) - 1 = (K_0 + w)C - F - z(q - 1)$$

saves a third gate when its two sides are compared directly. These are polynomial identities. The sparse-ring audit verifies each of the 22 residuals against both the predecessor's expression and its literal sparse coefficient data after substitution.

Given a new positive solution, $S = q^2 > 1$ implies $L_{\text{old}} > 0$, while $P_{\text{old}} = T+k > 0$ and $z_{\text{old}} = z+1 > 0$. The identities give a positive predecessor solution. Conversely, on a positive predecessor solution, $S = L_{\text{old}} + 1 = q^2$, and

$$a - E = sN > 0, \qquad c - \eta = k(a - E) > 0, \qquad T = P_{\text{old}} - k = E(c - \eta) > 0.$$

For the final shift, the inherited Pell/compiler theorem supplies $C > 0$, $X = wq = 2^R$, $R > 3q$, and $F < q$. Consequently $w > q > F$, and the transport equation gives

$$(z_{\text{old}} - 1)(q - 1) = (K_0 + w)C - F > 0.$$

Thus $z_{\text{old}} \geq 2$, so the new z is positive. This positivity step is a use of the inherited arithmetic theorem, not an inference from a finite test of a few tuples. The transformations are inverse on the positive zero sets.

Theorem 4.1 (The low-degree construction)**.** *ZFC proves that the 22-equation system above has no positive integer solution if and only if* Con(ZFC)*. Equivalently, ZFC proves the same statement for the quartic* $Q_{31} = 0$*.*

Proof. The shifts give the proved bijection to package 30's positive quartic system. The complete coefficient replacement theorem permits the new tuple in that system. Its inherited Pell/compiler equivalence and the literal K215 calibration give the theorem. □

4.2 Cost and the quartic used by the Jones branch

The system has 62 live binary gates: 33 multiplications and 29 additions or subtractions. The sum-of-squares finalizer adds 22 residual subtractions, 22 squares, and 21 additions, giving 127 gates. Its expansion has 297 monomials when the four coefficient ports remain independent formal symbols; the predecessor has 306 under that convention. This count does not identify algebraic coefficient ports with quantified unknowns.

Replacing every positive variable by its nonnegative successor gives a normalized quartic: at all positive variables equal to one the residual $E - X(a - E)$ equals one independently of the coefficient tuple. Thus the shifted quartic is nonzero at the all-zero tuple, as required by Jones. To keep the index coefficients and their exact source map simple, the signed construction below uses the predecessor's literal 31-variable quartic, rebound to the same new tuple. It has the same existential meaning. Its separate source data are included, and are not confused with the new 297-term quartic.

5 Two improved twelve-positive-variable constructions

There are two different improvements. The economical construction evaluates exactly package 30's degree-1081 polynomial in its four coefficient ports in 108, instead of 111, binary arithmetic operations. Its three-equation system costs 104 operations, or 103 after a justified shift of the positive transport coordinate. The second construction changes the polynomial: it has degree 1065 in the same twelve positive unknowns, compared with the previous degree record 1069. It costs 118 operations; its three-equation versions cost 114 and 113 operations, with maximum residual degree 1056. The latter combines package 24's sharp weight, package 25's exclusion of the negative main-root branch, and package 30's quadratic-remainder norm. The smaller carrier permitted by the branch exclusion creates two additional leading cancellations. These are comparisons with the frozen repository collection, not assertions of a global minimum.

All results in this section use one complete coefficient tuple satisfying the native blank-compiler interface. The fixed ports are $\beta = B - 1, K_0, M_C, M_F$, with M_F the shifted high mask. They are fixed integers, not additional unknowns. The coefficient construction elsewhere in this report supplies the tuple. Changing only one port is not justified.

5.1 Common polynomial definitions

The supplied variables are exactly

$$J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, t \in \mathbb{Z}_{>0}.$$

Every other displayed symbol denotes a polynomial expression in these variables. Put

$$\begin{aligned}
&q = \beta J + 1, && C_w = q - F - \alpha, && X = wq, && Y = sq^3, && E = XY,\\
&k = \eta + \zeta, && c = kY + \eta, && a = E + Y, && H = 4a + 3, && \Delta = a^2 + H,\\
&\ell = EkY, && A = 1 + \ell(\ell + k), && G = 1 + \Delta c^2, && b_0 = X + ac.
\end{aligned}$$

In particular $\Delta = (a+2)^2 - 1$. The unchanged index and transport expressions are

$$\begin{aligned}
R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,\\
N_k &= k - hE - R,\\
N_t &= (K_0 + w)C_w + q - F - z(q-1).
\end{aligned}$$

Define the strong and auxiliary expressions

$$\begin{aligned}
&I = ic^2, && D = 1 + \Delta I^2, && Q = \Delta^2 I^2, && d = Q - 1,\\
&v = cT, && b = c + RD, && e = Q(Dv^2 + b^2) - 1, && B_a = 2Qvb.
\end{aligned}$$

Notice that $Q = (\Delta I)^2$ is the square needed for rational Pell saturation; it is not an arbitrary positive auxiliary parameter.

Select one of two pairs (M, W), in both cases putting $L = 1 + HM$:

$$M_{\mathrm{econ}} = k(I^2 + T^2), \qquad W_{\mathrm{econ}} = b_0 + Ht, \tag{15}$$

$$C_* = E + 2(X + Y) + sq^2,$$

$$M_{\mathrm{canc}} = I\big(2T + RC_* ic\big), \qquad W_{\mathrm{canc}} = Lb_0 + Ht. \tag{16}$$

For a formal variable x let

$$\begin{aligned}
C(x) &= d(x^2 + M^2 A) - e,\\
F_*(x) &= \big[C(x)^2 + DB_a^2 - 4d^2x^2M^2A\big]^2 - 4DB_a^2C(x)^2,\\
F_H(u) &= H^8 F_*(u/H),\\
N &= F_H(W - L\sqrt{G})F_H(W + L\sqrt{G}).
\end{aligned}$$

The last expression is an integer polynomial. Its radicals describe a finite conjugate product and are not part of the arithmetic language or the list of quantified unknowns. A complete division-free evaluator is given below. The product equation and unshifted system are

$$P = N_k N_t(1 + N) - 1 = 0, \qquad \text{and} \qquad N_k = 1, \quad N_t = 1, \quad N = 0. \tag{17}$$

5.2 A cheaper exact norm evaluator

The following scalar formulas define N without radicals, division, or resultants. They are also the basis of the literal JSON gate lists:

$$\begin{aligned}
m &= HM, & n &= m^2A, & d_n &= dn, & Z &= H^2Q,\\
P_H &= ZDv^2, & U_H &= Zb^2, & e_H &= P_H + U_H - H^2, & j &= 4P_HU_H,\\
x_0 &= d(W^2 + L^2G), & \mathcal{T} &= 2x_0, & \kappa &= \left[d(W^2 - L^2G)\right]^2,\\
\mathcal{A} &= 2(d_n + e_H), & \mathcal{B} &= (d_n - e_H)^2 - j,\\
a_r &= \mathcal{T} - \mathcal{A}, & b_r &= \mathcal{B} - \kappa, & c_r &= 16d_nj,\\
v_r &= a_r(a_r\mathcal{T} + 2b_r) - c_r, & u_r &= b_r^2 - a_r^2\kappa,\\
N &= u_r(u_r + \mathcal{T}v_r) + \kappa v_r^2.
\end{aligned}$$

Here $d_n = dn$ means the product of d and n; it introduces no new parameter. In the emitted circuit $v^2 = c^2T^2$, so the separately useful T^2 is computed once.

To prove the evaluator, the elementary identity

$$(C^2 + j - K)^2 - 4jC^2 = (C^2 - K - j)^2 - 4Kj$$

rewrites $F_H(u)$, with $x = du^2$, as

$$(x^2 - \mathcal{A}x + \mathcal{B})^2 - c_rx.$$

For $u = W \pm L\sqrt{G}$, the two corresponding x values have sum $\mathcal{T}$ and product κ. Reduction modulo $x^2 - \mathcal{T}x + \kappa$ first replaces $x^2 - \mathcal{A}x + \mathcal{B}$ by $a_rx + b_r$, and then replaces its square minus c_rx by $u_r + v_rx$. Its norm is exactly $u_r^2 + \mathcal{T}u_rv_r + \kappa v_r^2$. This argument works when the radicands are squares or zero and when conjugates coincide; no discriminant is divided out.

Three reassociations each save one multiplication relative to package 30: compute the common factor H^2Q before forming P_H, U_H; compute κ by multiplying before squaring; and use the displayed Horner form for the final norm. These are identities over every commutative ring. At the same coefficient tuple, the economical polynomial is therefore unchanged, including all its integer zeros. A different complete tuple produces a different fixed polynomial through the same compiler interface. The sharper construction changes M, L, W as stated; its new degree is proved below.

5.3 Weak signs and the inherited interface

We use three precisely identified consequences of the blank-compiler theorem. First, the weak index/transport bootstrap, valid for $N_k = N_t = \epsilon \in \{1, -1\}$, supplies

$$\begin{gathered}
C_w \geq 0, \quad q \geq 16, \quad X \geq q, \quad Y \geq q^3,\\
3q < R < q^4 - 2, \quad E \geq q^4 > R + 2, \quad k = R + \epsilon + hE > E, \quad kY < c < k(Y + 1).
\end{gathered} \tag{18}$$

Second, restoring the normalized first, main, strong and auxiliary Pell equations, together with the main congruence, permits the inherited signed auxiliary-quotient soundness theorem. It forces $\epsilon = 1$ and extracts a blank K215 halt. Third, every blank halt has a canonical positive normalized witness. Those canonical witnesses satisfy $C_w > 0$ and $X = 2^R$. These are semantic compiler dependencies, not consequences of checking four integers modulo a few small primes.

The hypotheses of the weak bootstrap follow from a root of the product equation before any radical is recovered. Indeed

$$\kappa = x_0^2 - 4d^2W^2L^2G, \qquad N \equiv (u_r + x_0v_r)^2 \pmod 4.$$

At a zero of P, the three integer factors $N_k, N_t, 1+N$ are units. The last cannot equal -1, since this would give $N = -2$, which is not a square modulo 4. Thus $N = 0$ and $N_k = N_t = \epsilon$. For a solution of the system, $\epsilon = 1$ directly. No consistency assumption is used.

5.4 All branches and the sharp separation bound

Factoring the finite norm shows that $N = 0$ supplies coherent roots g, τ, f, y, initially in a finite algebraic extension, with

$$\begin{gathered} g^2 = G, \quad \tau^2 = A, \quad f^2 = D, \\ dy^2 = e - B_a f = Q(vf - b)^2 - 1, \\ W = Lg + H(M\tau + y). \end{gathered} \tag{19}$$

Conversely any such coherent choice makes a factor of N zero. This factorization neither divides by a radical nor presupposes the rationality of any root. The bounds that follow apply to every complex conjugate of y; such a conjugate can be imaginary.

For the economical weight, put $A_0 = a + 2$. From (18), $A_0 < 2E$, $c < 2kY$, $D < A_0^2I^2 < 4E^2I^2$, and $c < I^2$. Every auxiliary branch satisfies

$$|y| < 2(v\sqrt{D} + b).$$

The two elementary estimates

$$2v\sqrt{D} < 8kEYIT < \tfrac{1}{2}M_{\text{econ}}kY^2, \qquad 2b < 10RE^2I^2 < \tfrac{1}{2}M_{\text{econ}}kY^2$$

use $I^2 + T^2 \geq 2IT$, $kY > 8E$, $k > E$, and $Y^2 > 20R$. Moreover

$$\sqrt{A} > \ell = EkY > kY^2, \qquad \sqrt{G} - \sqrt{A} > k(Y^2 + Y - 1) > kY^2.$$

For the second inequality use $\sqrt{G} > (a+1)c > (E + Y + 1)kY$ and $\sqrt{A} < \ell + k$. We obtain the two uniform bounds

$$|y| < M_{\text{econ}}\sqrt{A}, \qquad |M_{\text{econ}}\tau + y| < M_{\text{econ}}\sqrt{G}. \tag{20}$$

For the cancellation weight a tighter, differently shaped estimate is needed. Let $S = \Delta I$, so $Q = S^2$. The weak bootstrap implies $\Delta > R + 2$ and $I^2 \geq c^4$, hence

$$Q - b = \Delta I^2(\Delta - R) - c - R > 0.$$

Consequently $b \leq d$ and $\sqrt{D} < S < d$. For $Z_1 = v\sqrt{D} + b > 1$, every auxiliary branch satisfies

$$|y|^2 \leq \frac{QZ_1^2 + 1}{d} < \left[Z_1\left(1 + \frac{1}{d}\right)\right]^2.$$

The strict inequality follows on subtracting the left member from the right: the numerator is $QZ_1^2 - d > 0$. Therefore

$$|y| < Z_1 + v + 1 < cT((a+2)I + 1) + c + RD + 1. \tag{21}$$

Set $\rho = (1 + 1/Y)/E$. Because $c/\ell < \rho$ and $(a+3)\rho < 2$, the T term in (21), divided by $\sqrt{A} > \ell$, is less than $2TI$. The remaining large term is controlled by the exact rational identity

$$(a+2)^2\rho = E + 2Y + X + 6 + \frac{Y}{X} + \frac{5}{X} + \frac{4}{Y} + \frac{8}{E} + \frac{4}{EY}.$$

Here $Y/X = sq^2/w \le sq^2$ and $5/X + 4/Y + 8/E + 4/(EY) < 1$. Since $X \ge 16$, this gives $(a+2)^2\rho < C_* - 2$. Also $(c+1)/\ell < 2$. Thus

$$\frac{|y|}{\sqrt{A}} < 2TI + R(C_* - 2)i^2c^3 + 2 \le 2TI + RC_*i^2c^3 = M_{\mathrm{canc}}.$$

We have proved, without recovering any square root,

$$|y| < M_{\mathrm{canc}}\sqrt{A}, \qquad |M_{\mathrm{canc}}\tau + y| < 2M_{\mathrm{canc}}\sqrt{A}. \tag{22}$$

Finally, the same weak bounds imply $G > A$ and $b_0 > \sqrt{A}$. Indeed $c > kY$ and $\Delta > E^2 + 2EY$ give

$$\Delta c^2 > k^2(E^2Y^2 + EY) = A - 1.$$

For the second inequality $b_0 > akY$ and

$$a^2k^2Y^2 - A = k^2\{EY(2Y^2 - 1) + Y^4\} - 1 > 0.$$

5.5 Why the smaller carrier is sound

In the economical construction, the two arguments of F_* associated with the two main roots are $(W \pm L\sqrt{G})/H$. If G were nonsquare, the vanishing of one factor of its rational quadratic norm would force the other to vanish as well. But the two arguments have distance $2L\sqrt{G}/H > 2M\sqrt{G}$, whereas each root of F_* has modulus less than $M\sqrt{G}$ by (20). This contradiction proves that G is a square. The selected g is positive: if it were negative then (19) would give $W < -L\sqrt{G} + HM\sqrt{G} < 0$, contrary to $W = b_0 + Ht > 0$.

For the cancellation construction, the weighted carrier makes the same conclusion possible with only $L = 1 + HM$, instead of package 24's $1 + 2HM$. The branch $g = -\sqrt{G}$ would require

$$x = \frac{W + L\sqrt{G}}{H} = t + \frac{L}{H}(b_0 + \sqrt{G}) > 2M\sqrt{A},$$

because $b_0 > \sqrt{A}$, $\sqrt{G} > \sqrt{A}$, and $L/H > M$. This contradicts (22). Thus the negative branch is impossible. If G were nonsquare, rational conjugacy would force that branch to vanish whenever the positive branch did, again a contradiction. Hence $g = \sqrt{G}$ is a positive integer.

In either construction, work in a finite normal splitting field containing the roots in (19). Now g is rational. If τ were irrational, an automorphism would send it to $-\tau$ while fixing g. Subtracting the two linear relations would require a change of magnitude $2M\sqrt{A}$ in y. Every conjugate of y has magnitude strictly less than $M\sqrt{A}$, an impossibility. Thus τ is rational, and hence an integer because $\tau^2 = A \in \mathbb{Z}$. The linear relation makes y rational, and $f = (e - dy^2)/B_a$ is rational because $B_a > 0$. Since $f^2 = D \in \mathbb{Z}$, f is integral.

Rational Pell saturation. If integers $S \geq 2, V$ and rational y satisfy $S^2V^2 - (S^2-1)y^2 = 1$, then y is integral. For completeness write $y = u/v_0$ in lowest terms, $v_0 > 0$. Its equation gives $v_0^2 \mid S^2 - 1$. The positive nonsquare integer $D_0 = (S^2-1)/v_0^2$ has the two integer-coordinate Pell solutions (S, v_0) and $(S|V|, |u|)$. Both coordinates in each pair are nonzero. Write these pairs as (χ_j, ψ_j) and (χ_l, ψ_l) in powers of the fundamental positive integer-coordinate Pell unit. The addition formulas give $\chi_{r+2j} \equiv -\chi_r \pmod{\chi_j}$ and $\chi_{2j-r} \equiv -\chi_r \pmod{\chi_j}$. Because $\chi_j \mid \chi_l$, reducing l modulo $2j$ and reflecting into $[0, j]$ shows that l is an odd multiple of j; strict growth excludes every index smaller than j. The Pell multiplication formulas then give $\psi_j \mid \psi_l$, so $v_0 \mid u$ and $v_0 = 1$. This uses the integer-coordinate Pell group, not a larger group of half-integral units.

Apply the lemma with $S = \Delta I$ and $V = vf - b$. It proves $y \in \mathbb{Z}$. Since $L \equiv 1 \pmod{H}$ and in both variants $W \equiv b_0 \pmod{H}$, the integral linear relation now yields $g \equiv b_0 \pmod{H}$. Therefore $\gamma = (g - b_0)/H$ is integral. It is also positive: $g > (a+1)c$ and $c > k > E > X$ imply $H\gamma = g - ac - X > c - X > 0$.

We have restored exactly

$$\begin{aligned} \tau^2 - \ell(\ell + k) &= 1, \\ (X + ac + H\gamma)^2 - \Delta c^2 &= 1, \\ f^2 - \Delta i^2 c^4 &= 1, \\ Q[c(Tf - 1) - Rf^2]^2 - (Q-1)y^2 &= 1, \qquad N_k = N_t = \epsilon. \end{aligned}$$

The auxiliary identity uses $f^2 = D$ and $vf - b = c(Tf - 1) - Rf^2$. Absolute values may be taken for τ and y. If $f < 0$, change the signs of f and T together, preserving Tf. The inherited signed-quotient compiler theorem applies, forces $\epsilon = 1$, and recovers a blank K215 halting computation.

Conversely, take a canonical positive normalized compiler witness, with $g = b_0 + H\gamma$ and positive τ, f, y. For the economical construction choose

$$t = \gamma + M(g + \tau) + y > 0;$$

for the cancellation construction choose

$$t = L\gamma + M\tau + y > 0.$$

The coherent factor then vanishes and $N_k = N_t = 1$. Every canonical witness can be extended in this way; no new restriction on its strong Pell index is imposed. This proves both directions of existential equivalence with the authentic compiler. Composing with the K215 syntactic-refutation calibration and negating existential solvability proves, within ZFC, that absence of positive integer solutions is equivalent to Con(ZFC).

5.6 The shifted systems

In the system only, replace the transport equation by

$$(K_0 + w)C_w = F + z(q - 1). \tag{23}$$

Here the new positive variable is related to the old one by $z_{\text{old}} = z + 1$. Every shifted solution therefore gives an unshifted solution, so soundness is immediate. For completeness use a canonical compiler witness, for which $C_w > 0$, $X = wq = 2^R$, $R > 3q$, and $F < q$. Consequently $w > q > F$, and the original transport equation gives

$$(z_{\text{old}} - 1)(q - 1) = (K_0 + w)C_w - F > 0.$$

Thus $z_{\mathrm{old}} \geq 2$ and $z = z_{\mathrm{old}} - 1$ is positive. This positivity is proved after restoring the compiler, rather than inserted into the weak bootstrap. Directly comparing the two sides of (23) saves one addition/subtraction. The product equations retain the unshifted variable.

5.7 Exact degrees, including the new cancellation

All four coefficient ports have degree zero and all twelve supplied variables have degree one. The economical norm is the unchanged package 30 polynomial, so its exact degree is 1072; its product degree is 1081. Its leading form, writing $q_0 = \beta J$ and $k_0 = \eta + \zeta$, is

$$(N_{\mathrm{econ}})_{1072} = 2^{32} i^{48} k_0^{96} w^{80} s^{176} q_0^{656} (q_0 - F)^{16}.$$

Here is a full upper and lower degree argument for the new construction. Replace every supplied variable by λ times itself, and use a subscript 0 for its highest homogeneous part. In the rational function field of the unscaled variables, the relevant square roots have formal Laurent expansions at $\lambda = \infty$. Their leading radicands are nonzero squares, so the ordinary formal binomial expansion constructs all branches without fractional orders. The basic degrees are

expression	E, H	Y	c	R	I	d	b	A, G	M	L
degree	6	4	5	4	11	46	38	22	27	33.

Define three nonzero leading forms

$$\mathfrak{B} = R_0 E_0^2 i^2 c_0^4, \qquad \mathfrak{D} = R_0 E_0 Y_0 i^2 c_0^4, \qquad \mathfrak{C} = R_0 E_0 (s q_0^2) i^2 c_0^4.$$

Their degrees are $38, 36, 35$, respectively.

The first radical satisfies $\sqrt{A} - \ell = O(\lambda)$. For every f branch the auxiliary roots satisfy

$$y = \varepsilon(b - vf) + O(\lambda^{-8}), \qquad \varepsilon \in \{1, -1\}, \qquad \deg(vf) = 23.$$

The exact identity

$$C_* \ell - \Delta c = (2EX + E s q^2 - Y^2 - 4E - 4Y - 3)c - E C_* \eta$$

has unique highest homogeneous term $E_0(s q_0^2) c_0$, of degree 14. Consequently

$$M\sqrt{A} - b = \mathfrak{C} + \text{terms of order below } 35. \tag{24}$$

Indeed the difference is $R i^2 c^3 (C_* \ell - \Delta c) + O(\lambda^{28})$.

Let $g_+ = \sqrt{G}$ and $\gamma_\pm = (\pm g_+ - b_0)/H$. The small quotient root γ_+ has order -1, hence $L\gamma_+$ has order 32. Also $g_+ = ac + O(\lambda^5)$ and

$$g_+ + b_0 - 2\sqrt{A} = 2Y_0 c_0 + \text{terms of order below } 9.$$

Multiplying by M, whose top form is $R_0 E_0 i^2 c_0^3$, gives

$$M(g_+ + b_0 - 2\sqrt{A}) = 2\mathfrak{D} + \text{terms of order below } 36. \tag{25}$$

The exact sixteen-branch factorization is

$$N = d^8 H^{16} \prod (t - y - M\tau - L\gamma), \tag{26}$$

where the coherent choices of g, τ, f, y are all included with multiplicity. For γ_+, four factors have degree 35 with signed top forms $\pm\mathfrak{C}$; the other four have degree 38 with signed top forms $\pm 2\mathfrak{B}$. For γ_-, the two branches with $\tau = +\sqrt{A}$ and $y = +(b - vf) + O(\lambda^{-8})$ have degree 36 and top form $2\mathfrak{D}$, by (24) and (25). The remaining six have degree 38: four top forms are $2\mathfrak{B}$ and two are $4\mathfrak{B}$. This exhausts every branch. It proves the degree upper bound, not just a degree observed on a numerical specialization. Multiplying the nonzero top forms gives

$$\begin{aligned}\deg N &= 8 \cdot 46 + 16 \cdot 6 + 4 \cdot 35 + 2 \cdot 36 + 10 \cdot 38 = 1056,\\ N_{1056} &= 2^{14} d_0^8 H_0^{16} \mathfrak{B}^{10} \mathfrak{C}^4 \mathfrak{D}^2\\ &= 2^{46} i^{48} k_0^{96} w^{74} s^{176} q_0^{646} (q_0 - F)^{16}.\end{aligned}$$

This form is nonzero for every fixed $\beta > 0$. Since

$$(N_k)_7 = -hwsq_0^4, \qquad (N_t)_2 = w(q_0 - F - \alpha) - zq_0,$$

the product has exact degree $1056 + 7 + 2 = 1065$. For example its coefficient of

$$J^{667} hz\, i^{48} \eta^{96} w^{75} s^{177}$$

is $2^{46}\beta^{667} > 0$. The three-equation residual degrees are exactly $7, 2, 1056$; shifting z by one preserves these leading degrees.

5.8 Literal cost, checks, and proof status

The supplied evaluator charges every binary $+, -, \times$, including one by a fixed numeral. Intermediate expressions are shared; comparisons in systems are free under the repository convention.

Construction	Equations	Degree	Multiplications	Total operations
Economical product	1	1081	65	108
Economical system	3	1072	63	104
Economical shifted system	3	1072	63	103
Cancellation product	1	1065	72	118
Cancellation system	3	1056	70	114
Cancellation shifted system	3	1056	70	113

The checks compare each new norm with an independently implemented eight-dimensional multiquadratic conjugate product on 50 signed integer assignments, check the parity and transport identities on those assignments, and reject dead gates or unused ports. Full univariate coefficient lists over two primes confirm the cancellations and the stated leading forms; this is stronger than merely propagating degree upper bounds through the gates. The mathematical arguments above, rather than those finite examples, establish the unbounded root-recovery theorem and uniform exact degrees. No new end-to-end Lean or Rocq check is claimed, and no assertion that ZFC is consistent is made.

6 Sixteen unrestricted integers by a cofinal square radix

The new integer construction combines three components that were previously separate: package 30's compact evaluation of the Jones ratio for its 31-variable quartic; package 29's signed relation combiner; and a new restriction of the coding radix to a cofinal family of scaled squares. The last

step removes one positivity witness. A smaller signed separator also lowers the degree and evaluation cost. The result is a completely specified polynomial in *sixteen unrestricted integers*, with

$$\deg P_{16} = 813056\,5^{32} + 10024, \qquad \operatorname{ops}(P_{16}) = 1865.$$

These are improvements over the delivered 17-integer circuit, not a new general bound for arbitrary recursively enumerable sets. In particular, Sun's 2026 ten-integer theorem is a different, more general representation theorem, whose compiler is not instantiated here.

6.1 Fixed index and literal Jones expressions

Use the literal 31-variable quartic of package 30, supplied in the archive as inert source data. Its residuals are quadratic; shifting every positive coordinate by one and summing their squares gives a normalized polynomial $P_{31}(x, \mathbf{n})$, independent of the dummy input x. The source's fifth residual is $EX + E - aX$, which equals one at the all-ones tuple, so $P_{31}(x, \mathbf{0}) > 0$. Its coefficient of the fourth power of the retained first Pell root is one; hence its degree is exactly four. The quartic is rebound to the *whole* coefficient tuple proved in the coefficient section; its underlying four-port interface is unchanged. This is a semantic use of that interface theorem, not an inference from matching file hashes.

Put $\nu = 31$ and $L = 5^{32}$. If p_α is the coefficient of the exponent vector $\alpha \in \mathbb{N}^{32}$, $|\alpha| \le 4$, in the shifted quartic, define

$$c_\alpha = (4 - |\alpha|)! \prod_{r=0}^{31} \alpha_r!\, p_\alpha, \qquad e(\alpha) = \sum_{r=0}^{31} \alpha_r 5^r.$$

There are 402 nonzero coefficient records, each a specified polynomial in the four fixed compiler ports. The bit lengths of those ports, in order (β, K_0, M_C, M_F), are

$$(1064921, 1070004, 1064921, 1064922).$$

For a coefficient record $c_\alpha = \sum_t a_t \prod_j C_j^{e_{tj}}$, the finite exact bound

$$|c_\alpha| \le H_\alpha := \sum_t |a_t| 2^{\sum_j e_{tj} \operatorname{bits}(C_j)}, \qquad H_\alpha + 1 < 2^{4269855}$$

is verified for every record by the supplied index certificate. Unlisted records have coefficient zero. Consequently the following are admissible fixed integers:

$$z_* = 2^{4269855}, \qquad e_*(T) = \sum_{|\alpha| \le 4} (z_* + c_\alpha) T^{L - e(\alpha)},$$

$$y_* = e_*(2z_*), \qquad b_* = (2z_*)^{L+1} = 2^{4269856(L+1)}.$$

The corresponding unused Jones index coordinate is $u_* = \sum_{r=1}^{31} (2z_*)^{5^r}$. Base-five uniqueness makes the exponents in e_* distinct, and every digit lies strictly between one and $2z_*$. Thus

$$2(2z_*)^L \le y_* < (2z_*)^{L+1} = b_*.$$

The integer y_* is needed to state the imported index theorem but *does not occur as a coefficient leaf of the new polynomial.* It is replaced there by the simple power-of-two coefficient b_*. None of these fixed integers is an existential parameter.

The eight base variables are

$$(e, g_0, h_0, s_0, w_0, \phi, i_0, j_0) \in \mathbb{Z}^8.$$

Set $g = g_0 + 1$, $h = h_0 + 1$, $s = s_0 + 1$, $w = w_0 + 1$, $i = i_0 + 1$, $j = j_0 + 1$, and define

$$\begin{aligned} b &= b_* e^2, & \mathcal{B} &= b^5, & q &= \mathcal{B}^L, \\ \ell &= \sum_{r=1}^{31} \mathcal{B}^{5^r}, & \lambda &= \sum_{r=0}^{4L-1} \mathcal{B}^r, & N &= q^{16}, \\ \theta &= \mathcal{B} - 2z_*, & G &= 32b\mathcal{B}^{5^{31}}. \end{aligned}$$

Write $E_* = e_*(\mathcal{B})$, and put

$$\begin{aligned} R = {} & \big[g + E_* q^3 + \ell q^5 + \{2(E_* - z_*\lambda)(1 + \mathcal{B} + g)^4 + \lambda\mathcal{B} + \lambda\mathcal{B}q^4\}q^7\big](N^2 - N) \\ & + \big[q^3 - b\ell + \ell + \theta\lambda q^3 + (\mathcal{B} - 2)q^8\big](N^2 - 1). \end{aligned}$$

Continue with the literal Jones ratio expressions:

$$\begin{aligned} U &= N^2 w, & Y &= N^2 s, & M &= RY, \\ P &= 2M^2 U, & K &= R + 1 + h(P - 1), & A &= M(U + 1), \\ B_1 &= 2R + 1, & W_1 &= bw, & C &= B_1 + W_1 + \phi, \\ J &= 4A - 5, & Q &= 3W_1 C - 2(W_1^2 - 1), & D &= (A^2 - 1)C^2 + 1, \\ E &= iJDC^2, & F &= (A^2 - 1)E^2 + 1, & G_1 &= A + F(CD + 1 - A), \\ H &= B_1 + jC, & I &= (G_1^2 - 1)H^2 + 1. \end{aligned}$$

Finally set

$$\begin{aligned} a_0 &= (P^2 - 1)K^2 + 1, & a_1 &= DFI, \\ D_0 &= JF, & T_0 &= QF + J(H - C), \\ m &= K^2 G - 4(C - KY)^2 G - 8gK^2. \end{aligned}$$

Every intermediate symbol is a polynomial abbreviation. In particular, all exponents are fixed natural numbers, and no radical or variable exponent occurs in the delivered arithmetic circuit.

6.2 The new cofinal-radix lemma

Lemma (restriction to scaled squares). For the fixed admissible Jones index, the source existential predicate

$$a_0, a_1 \in \mathrm{Sq}(\mathbb{Z}), \qquad D_0 \mid T_0, \qquad m > 0 \tag{JR}$$

is equivalent to the parent-quartic zero predicate when $b = b_* e^2$, $e \in \mathbb{Z} \setminus \{0\}$, and the other seven source coordinates are nonnegative.

Proof. Soundness is immediate from $b \geq b_* > y_*$: the integer $\varepsilon = b - y_* - 1$ is nonnegative, and replacing the usual source definition $b = y_* + 1 + \varepsilon$ by this value changes no ratio expression. The source soundness theorem applies literally.

For completeness, inspect the constructive proof of `Jones1982.USys_of_mem` in the pinned `Master3.lean`. For a fixed parent solution $\mathbf{n}$, its choice $b = 2^v$ uses the equality $v = xy_* + \sum_r n_r + 1$ only to establish the initial bounds $v \geq 1$, $xy_* < b$, and $n_r < b$. After these bounds, the entire construction uses only those inequalities, the positive power-of-two condition on b, and the fixed index. The displayed source was audited for every occurrence of that equality: there is no later use. Hence the same ordinary mathematical construction works for every integer $v \geq xy_* + \sum_r n_r + 1$.

Write $b_* = 2^\kappa$, where $\kappa = 4269856(L+1)$. Choose $t \geq 0$ with $\kappa + 2t$ exceeding this bound, take $v = \kappa + 2t$, and put $e = 2^t$. Then $b = 2^v = b_* e^2$. The unchanged `Nine.reduced_of_mem` construction and `Nine.ratio_complete` give the ratio witnesses. This is an explicit strengthening of the source construction; no claim that an arbitrary Diophantine witness may be replaced by a square is used. □

6.3 Five base guards and one margin guard

Only $\mathbf{x} = (g_0, s_0, w_0, \phi, j_0)$ needs direct nonnegativity guards. The variables h_0, i_0 remain signed. Introduce six integers $z_1, \ldots, z_6$. Put $C_0 = 1$ and

$$d_r = 3x_r + 2, \qquad B_r = 1 + 4d_r C_{r-1}^2 z_r^2, \qquad C_r = C_{r-1}B_r \quad (1 \leq r \leq 5),$$
$$d_6 = 3m - 1, \qquad B_6 = 1 + 4d_6 C_5^2 z_6^2, \qquad C_* = C_5 B_6.$$

Lemma (coprime Pell guard). If every $z_r \neq 0$, then

$$C_* \in \mathrm{Sq}(\mathbb{Z}) \quad \Longrightarrow \quad g_0, s_0, w_0, \phi, j_0 \geq 0, \quad m > 0.$$

Conversely these inequalities permit a choice of nonzero z_r for which C_* is a square. Moreover, for arbitrary signed base coordinates and nonzero z_r, $C_r \neq 0$ and $|C_*| \geq 3^6$.

Proof. Each $d_r \equiv 2 \pmod 3$ is nonzero, and each $B_r \equiv 1 \pmod 4$ is nonzero. Also $\gcd(B_r, C_{r-1}) = 1$, so the factors B_r are pairwise coprime. If their product is a square, unique factorization makes each $|B_r|$ a square. A negative square is zero or three modulo four; therefore every B_r is a positive square. A negative d_r, with both $C_{r-1}, z_r \neq 0$, would give $B_r \leq -3$. Thus $d_r > 0$, giving the asserted inequalities.

Conversely, each positive $d_r \equiv 2 \pmod 3$ is nonsquare. The Pell equation $u^2 - d_r v^2 = 1$ has a nontrivial positive solution whose v is divisible by any specified positive integer: reduce a Pell unit matrix modulo that integer and take a positive power equal to the identity in the finite matrix group. Take the modulus $2|C_{r-1}|$ and set $z_r = v/(2C_{r-1})$. This makes $B_r = u^2$ and $z_r \neq 0$. Induction proves completeness. Finally, without any positivity assumption, $|B_r| \geq 3$ when $z_r \neq 0$; multiply the six inequalities. □

Lemma (the two unguarded base coordinates). Assume $e \neq 0$, the five base guards are nonnegative, a_0 is a square, and $m > 0$. Then $h \geq 1$. If also $i \neq 0$, replacing i by $|i|$ preserves (JR) and makes $i_0 = |i| - 1$ nonnegative.

Proof. Here $b \geq b_* > y_*$, so $\mathcal{B} > 2z_*$, $G > 0$, and $g \geq 1$. From $m > 0$ obtain

$$K \neq 0, \qquad g < G/8 = 4b\mathcal{B}^{5^{31}}, \qquad 4(C - KY)^2 < K^2.$$

Write $r = 5^{31}$. The first two bounds imply $1 + \mathcal{B} + g < \mathcal{B}^{r+2} \leq q$. Also $\ell < \mathcal{B}^{r+1}$, so $b\ell < q^3$. In the first bracket defining R, the positive term $\lambda\mathcal{B}q^4$ strictly absorbs $2z_*\lambda(1 + \mathcal{B} + g)^4$; every remaining term is nonnegative. The second bracket is positive as well. Thus $R \geq N^2 - N \geq 2$. Now $Y, U \geq 1$, $C = 2R + 1 + bw + \phi > 0$, and $P = 2R^2Y^2U > R + 2$. If $K < 0$, then $|C - KY| > |K|$, contradicting the displayed strict inequality. Hence $K > 0$. If $h \leq 0$, then $1 \leq K \leq R + 1 < P$. The case $K = 1$ would imply $P - 1 \mid R$, impossible. For $2 \leq K < P$,

$$(PK - 1)^2 < (P^2 - 1)K^2 + 1 < (PK)^2,$$

contradicting squarehood of a_0. This proves $h \geq 1$. Finally i occurs only linearly in E, whose only relevant consumer is E^2 in F. Changing the sign of i therefore changes none of a_0, a_1, D_0, T_0, m. □

6.4 Divisibility and nonvanishing with one extra coordinate

Use the fixed intersective polynomial

$$\Pi(t) = (t^3 - t - 1)(t^2 + 23).$$

It has no rational root. It has a root modulo every positive integer: for odd primes other than 23, either the cubic has a simple root, or it is irreducible and its Frobenius three-cycle makes its discriminant -23 a nonzero square, giving a simple root of the quadratic. These roots lift to all prime powers. At 23, the cubic has the simple root 3; at 2, $-23 \equiv 1 \pmod 8$ has square roots modulo every power of two. The Chinese remainder theorem finishes the proof.

Define the homogeneous degree-five polynomial

$$\mathcal{R}(D, T, v) = ((T + Dv)^3 - D^2(T + Dv) - D^3)((T + Dv)^2 + 23D^2).$$

For integers $D \neq 0, T, Z$,

$$(D \mid T \text{ and } Z \neq 0) \quad \Longleftrightarrow \quad \exists v \in \mathbb{Z} : \ ZD^5 \mid \mathcal{R}(D, T, v). \tag{IM}$$

Indeed $\mathcal{R} = D^5\Pi(T/D + v)$ never vanishes, since $T/D + v$ is rational. If $D \mid T$, intersectivity supplies v so that $Z \mid \Pi(T/D + v)$. Conversely, if a prime satisfies $\alpha = v_p(D) > \beta = v_p(T)$, then $v_p(T + Dv) = \beta$, and the two factors in $\mathcal{R}$ have unique lowest-valuation terms of valuations 3β and 2β. Hence $v_p(\mathcal{R}) = 5\beta < 5\alpha$, contradicting divisibility by D^5. The case $Z = 0$ is excluded by nonvanishing.

For the Jones expressions, $D_0 = JF \neq 0$ at *every* integer assignment: $J = 4A - 5 \neq 0$; and if $F = 0$, then $(1 - A^2)E^2 = 1$ gives $A = 0, E = \pm 1$, although $E = iJDC^2$ would then be divisible by five. This contradiction does not use any guard or desired square condition.

6.5 The literal sixteen-variable polynomial

Besides the eight base coordinates and six z_r, introduce $v, u \in \mathbb{Z}$. These are all sixteen unknowns. Put

$$\begin{aligned}
Z &= ie \prod_{r=1}^{6} z_r, & S &= ZD_0^5, & T &= \mathcal{R}(D_0, T_0, v),\\
w_1 &= C_*, & w_2 &= C_*(P^2 + K^2 + 2), & X &= Su - T,\\
A_0 &= S^2 C_*, & A_1 &= S^2 w_1^2 a_0, & A_2 &= S^2 w_2^2 a_1,\\
V &= X^2 + A_2 - A_0 - A_1, & W &= V^2 + 4X^2 A_2 - 4A_0 A_1.
\end{aligned}$$

The equation is

$$\boxed{P_{16} = W^2 - 16V^2X^2A_2 = 0.}$$

Lemma (signed three-root finalizer). A zero of this polynomial gives

$$S \neq 0, \qquad S \mid T, \qquad C_*, a_0, a_1 \in \mathrm{Sq}(\mathbb{Z}). \tag{F}$$

Conversely, these conditions on the earlier coordinates allow an integer choice of u making the polynomial zero.

Proof. If $S = 0$, then $A_0 = A_1 = A_2 = 0$, $X = -T$, and $P_{16} = T^8 \neq 0$. Thus any zero has $S \neq 0$, so $i, e, z_1, \ldots, z_6$ are nonzero. The guard lemma, before any squarehood conclusion, gives $|C_*| \geq 3^6$. Consequently

$$|w_1| > \sqrt{|C_*|}.$$

For arbitrary real P, K,

$$\sqrt{|(P^2-1)K^2+1|} \leq P^2 + K^2 + 1.$$

To verify this, first bound the absolute radicand by $(P^2+1)K^2+1$, then subtract this expression from $(P^2+K^2+1)^2$; the result is $P^4 + K^4 + P^2K^2 + 2P^2 + K^2 \geq 0$. It follows that

$$|w_2| > \sqrt{|C_*|} + |w_1|\sqrt{|a_0|}.$$

These estimates permit signed C_*, w_1, w_2; positivity has not been assumed.

Choose complex square roots $r_0^2 = C_*$, $r_1^2 = a_0$, $r_2^2 = a_1$. Direct multiplication gives the polynomial identity

$$P_{16} = \prod_{\epsilon_0,\epsilon_1,\epsilon_2 \in \{-1,1\}} (X - S(\epsilon_0 r_0 + \epsilon_1 w_1 r_1 + \epsilon_2 w_2 r_2)).$$

At a zero, one of these signed sums is rational. Under any automorphism of the finite Galois extension generated by the three roots, each root changes only by sign. If the largest nonzero root whose sign changes has index 1 or 2, its weighted modulus strictly exceeds the sum of all earlier weighted moduli by the preceding bounds: a nonzero root of an integer radicand has modulus at least one. A change only at index 0 is impossible as well. Thus an automorphism cannot preserve the rational sum while changing a nonzero root. Every root is fixed, hence rational, and a rational square root of an integer is an integer. This handles negative candidate radicands, zero roots, and dependent square classes. The resulting integral signed sum equals $u - T/S$, so $S \mid T$. This proves necessity. Conversely, select integral roots and take $u = T/S + r_0 + w_1 r_1 + w_2 r_2$. A factor vanishes. □

Theorem (the concrete integer equation). With the fixed coefficients above,

$$\mathrm{ZFC} \vdash (\nexists \mathbf{t} \in \mathbb{Z}^{16} : P_{16}(\mathbf{t}) = 0) \longleftrightarrow \mathrm{Con(ZFC)}.$$

Proof. A zero first gives (F). Apply (IM) to recover $D_0 \mid T_0$ and all nonzero conditions. The coprime guard gives the five nonnegative base coordinates and $m > 0$. The unguarded-coordinate lemma gives $h_0 \geq 0$ and permits replacing i by $|i|$. Because $e \neq 0$, the derived source coordinate $\varepsilon = b_* e^2 - y_* - 1$ is nonnegative. We have therefore recovered the literal source Jones predicate, which gives a zero of the parent quartic and hence, through the whole-tuple compiler and the ordinary K215 calibration, a finite ZFC refutation.

Conversely, a finite refutation gives a parent-quartic zero. The cofinal-radix lemma gives Jones witnesses with $b = b_* e^2$, $e \neq 0$. The Pell guard supplies the six nonzero z_r and a square C_*. Intersectivity supplies v with $S \mid T$, and the finalizer supplies u. All steps are ordinary number theory and finite Galois theory formalizable in ZFC. Negating this two-way existential equivalence proves the assertion. It proves neither consistency nor inconsistency. □

6.6 Exact complexity and reproducible checks

The square-radix substitution doubles the degree contributed by the radix coordinate, while removing one guard. With $L = 5^{32}$, selected exact degrees are

R	$470L$	P	$1900L + 3$
K	$1900L + 4$	m	$4442L + 12$
a_0	$7600L + 14$	a_1	$49280L + 68$
D_0	$13750L + 20$	C_*	$4442L + 1103$
S	$68750L + 108$	T	$68750L + 105$
X	$68750L + 109$	w_2	$8242L + 1111$
A_2	$203264L + 2506$	P_{16}	$813056L + 10024$

The degree of A_2 strictly exceeds those of X^2, A_0, A_1. Therefore V has leading form A_2, W has leading form A_2^2, and the leading form of the final output is A_2^4. On the diagonal specialization this is a positive integer times a power of the positive fixed coefficient b_*. The supplied leading-term certificate keeps that power symbolic, establishing the stated exact degree for the authentic tuple rather than only a generic upper bound.

All fixed powers are expanded into binary multiplications; squaring and scalar multiplication are charged. Reuse is free, and fixed coefficient construction is excluded from variable-evaluation cost. The emitted, fully live DAG has 705 additions, 26 subtractions, and 1134 multiplications, totaling 1865. Its source digit polynomial uses the compact Horner evaluator and shared radix-five powers of package 30; no imported Python program is executed.

The independent diagnostics evaluate actual fixed coefficients modulo three primes and compare 90 signed tuples both with a direct formula and with the unchanged inherited Jones ratio after substituting $\varepsilon = b_* e^2 - y_* - 1$. They also check 200 independent eight-factor norm products on integral roots and another 200 products in the rank-eight free radical algebra with arbitrary signed radicands, the basic separator inequalities on 57,222 assignments, and intersectivity modulo every integer through 1200. These finite tests check definitions and arithmetic. The unbounded correctness argument is the proof above, together with the explicitly identified source compiler and Jones theorems; the archive does not claim a new end-to-end Lean or Rocq certificate.

Seventeen-integer companion. Retaining $b = y_* + 1 + \varepsilon$, guarding all six base coordinates as in package 29, and using the improved separator $w_1 = C_*$ gives a companion polynomial in seventeen integers. Its exact degree is $406528\,5^{32} + 27512$, and its 1874 live operations consist of 709 additions, 26 subtractions, and 1139 multiplications. It uses the same new coefficient tuple and index. The corresponding proof is the signed-three-root proof above with seven guard factors and without the cofinal-square-radix step. This is an explicit alternative when degree is valued more highly than the final integer-variable reduction.

7 Verification, archive contents, and remaining questions

7.1 A reproducible finite audit

The archive is self-contained for the new finite calculations. Python 3.10 or later and its standard library suffice; no network connection, symbolic algebra package, proof-assistant installation, or machine simulation is needed. From the extracted top-level directory run

```
python3 reproduce.py
```

The runner authenticates the included source files, rebuilds the four coefficient integers, runs an independent reconstruction of that tuple, rebuilds the low-degree and positive circuits, rebuilds both signed circuits and their fixed-index certificates, and runs the independent signed audit. It compares the regenerated literal mathematical artifacts with the delivered versions and writes a receipt under `verification/`. The source map includes SHA-256 and pinned Git blob identities. Imported programs are retained as reference text; the new builders read their JSON data and do not execute the imported programs.

The principal checks have the following scopes.

1. **Coefficient construction.** The 430 literal actions determine the 943 ordered tiles. Both implementations check 69230 native-basis convolution identities, clause-band separation, anchor synchronization, zero padding fields, carry bounds, population balance, the general padding controller's orders and valuation, and its unit table. All four raw byte strings match an independent reconstruction.

2. **Quadratic and quartic algebra.** An exact sparse polynomial ring verifies all 22 residual identities after the coordinate shifts, against both the predecessor's expressions and its literal sparse data. It also verifies the comparison circuit, the sum of squares, degree four, the coefficient of τ^4, normalization, and gate liveness.

3. **Positive norm algebra.** For each retained family, 50 exact conjugate products in a free radical algebra agree with the scalar evaluator; 50 parity checks and 50 transport identities also pass. Full univariate coefficient calculations over each of two prime fields verify the degree and leading-form predictions. The Laurent argument in the proof supplies the universal upper bound and nonzero leading form; the modular calculations are independent implementation diagnostics.

4. **Signed algebra and fixed index.** Each signed candidate has an exact symbolic leading-term certificate, live-input/gate checks, 90 modular comparisons at three primes using the actual coefficient tuple, 200 norm products with integral roots, and 200 further norm products with arbitrary signed radicands. There are 57222 finite separator checks and intersectivity checks through modulus 1200. The separate signed reviewer checks another 300 arbitrary signed-radicand products and 300 guard stacks, and reads the circuit as inert JSON.

Separate proof reviews examine the infinitary steps that finite arithmetic cannot settle: the new padding modulus, the sharp positive-root bound, rational Pell saturation, the cofinal Jones radix, and the order of the signed nonvanishing and separation arguments. These reviews were performed by separate agents within this AI-assisted collaboration; they are not external human endorsement or proof-assistant certification.

7.2 Which files define the equations

Location	Role
`report.tex`, `sections/`	Full mathematical exposition and proofs
`report.pdf`	Compiled, visually inspected report
`coefficients/`	Literal machine, four actual integer payloads, finite recipes, binding, and receipts
`positive/`	Six live circuits: two single equations and their ordinary/shifted three-equation systems
`quartic/`	22-equation quadratic system, quartic sum of squares, and exact sparse data
`integers/generated/`	Sixteen- and seventeen-integer circuits, fixed-index recipes, and degree certificates
`sources/`	Authenticated inherited proof texts, endpoint statements, and source map
`reviews/`	Independent mathematical reviews and separately implemented audits
`RESULTS.json`	Explicit domain, degree, cost, and binding for every delivered form
`MANIFEST.sha256`	File integrity list for the delivered release

All named intermediate quantities in a circuit are abbreviations for polynomials, not additional quantified variables. Every fixed coefficient is either a delivered integer byte string or an exact finite integer recipe. No coefficient depends on choosing whether ZFC is consistent. No input port remains an unspecified free parameter in the fixed-instance bindings.

The two signed polynomials deliberately use package 30's old literal 31-variable quartic, with the complete new coefficient tuple. The new 62-operation system and its 297-term quartic are a separate improvement. Substituting the latter into the Jones branch would require regenerating the weighted index coefficients; it is not done implicitly here.

For the PDF, run `make pdf`, or run `pdflatex` twice on `report.tex`. The supplied Makefile uses no shell escape. The LaTeX sources use ordinary AMS, Latin Modern, geometry, microtype, table, listing, and hyperlink packages.

7.3 Interpretation and next useful reductions

The new results answer distinct versions of "simpler." The sixteen-integer construction improves the current explicit signed variable count in this repository. The twelve-positive-variable cancellation improves degree; the other twelve-variable presentation improves cost. The quadratic system gives a short low-degree form, and the coefficient adapter reduces the common direct coefficient payload. These conclusions do not establish an optimum in any unrestricted class of Diophantine presentations.

The most consequential unresolved extensions are a fully instantiated ten-integer construction based on Sun's different representation, further removal of signed positivity witnesses, and a kernel-checked proof of the new compiler adapter and finalizers. An improvement to the nine-positive- variable count or to the established 67-operation equation would require another idea; this report does not claim those improvements.

The result proves an equivalence. Assuming ZFC is consistent, Gödel's second incompleteness theorem prevents ZFC from proving that the equation has no solutions. ZFC can nevertheless prove that each solution encodes a finite refutation, and that every finite refutation supplies a solution.

References

[1] Vladimir Reshetnikov, *ProveIt*, frozen commit `8c6eec9d19ef80538c8e61ff6e0180040583446a`. Repository: https://github.com/VladimirReshetnikov/ProveIt. The bundled source map gives the precise path of each imported text, literal data file, and construction.

[2] ProveIt package `24-integer24`, *Smaller explicit Diophantine forms of ZFC consistency*. Especially `sections/coefficients.tex`, `sections/positive.tex`, and the coefficient generator.

[3] ProveIt package `25-hybrid12`, *Smaller Diophantine forms of ZFC consistency*. Especially `coefficients/PROOF.md`, `proofs/HYBRID12_PROOF.md`, and the inherited compiler notes.

[4] ProveIt package `27-wave4`, *Smaller Diophantine presentations of ZFC consistency*. Especially the Wang943 compiler/interface proofs and the retained transport-shift proof.

[5] ProveIt package `29-simpler-forms`, *Smaller explicit Diophantine forms of ZFC consistency*. Especially `signed17/SIGNED17_PROOF.md`.

[6] ProveIt package `30-quartic31`, *Smaller explicit Diophantine forms for Con(ZFC)*. Especially the literal quartic, compact Jones evaluator, and `positive/POSITIVE12_PROOF.md`.

[7] James P. Jones, *Universal Diophantine equation*, Journal of Symbolic Logic 47 (1982), 549–571. https://doi.org/10.2307/2273588.

[8] Zhi-Wei Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, 2 September 2026, Theorem 1.1. https://arxiv.org/abs/2609.01594v2.